



NCCER Module 22102



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Heavy Equipment Safety





Section Four

Trenching and Excavation Safety

Objective

4. Identify safety practices associated with trenches and excavations.
 - a. Explain how soil composition relates to trench failures.
 - b. Describe the methods used to protect workers in trenches and excavations.
 - c. Identify personal safety hazards related to dust and soils.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



4.0.0 Trenching and Excavation Safety (1 of 3)

- A trench is an excavation that is deeper than it is wide, and usually not wider than 15 feet (4.6 meters).
- Trenches are typically dug to create footings, lay pipe, or place manholes.
- If the walls are not properly secured by **shoring**, **sloping**, or **shielding**, they will collapse.



LOADING

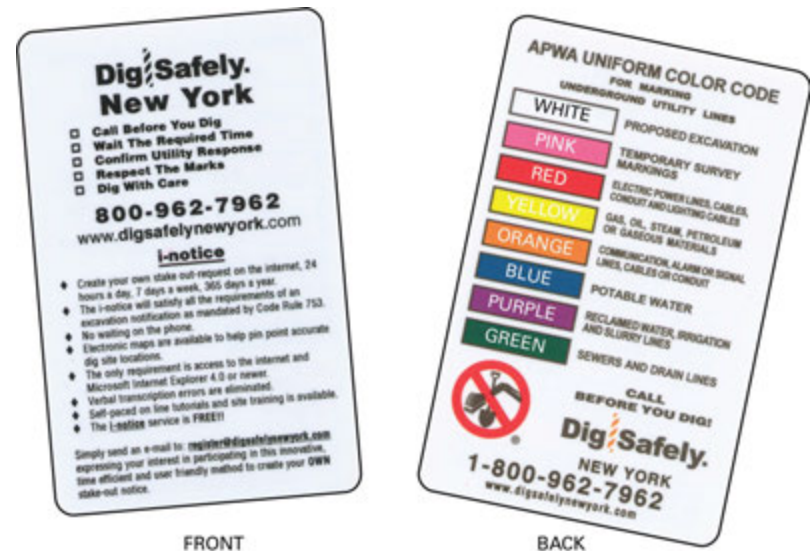


TRENCHING



4.0.0 Trenching and Excavation Safety (2 of 3)

- The collapse of unsupported trench walls can instantly crush and bury workers.
- Buried utilities in the excavation area must be located and marked.
- Each state has its own agency to coordinate the location of buried utilities.



4.0.0 Trenching and Excavation Safety (3 of 3)

When heavy equipment is used around trenches, safety precautions must be exercised at all times to prevent injury to yourself and others.

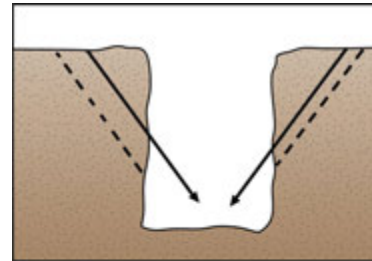
Hazards include:

- Cave-ins due to trench failure
- Flooding from broken water or sewer mains
- Electrical shock from striking electrical cable in the trench or striking overhead lines
- Toxic liquid or gas leaks from nearby facilities, pipes, or idling vehicles
- Falling dirt or rocks from an excavator bucket
- Oxygen-depleted environment

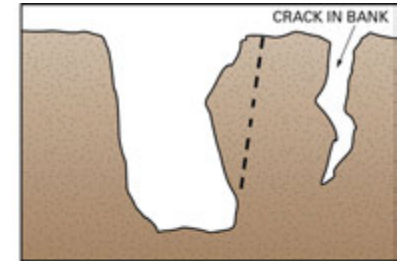


4.1.0 Soil Composition and Trench Failures (1 of 2)

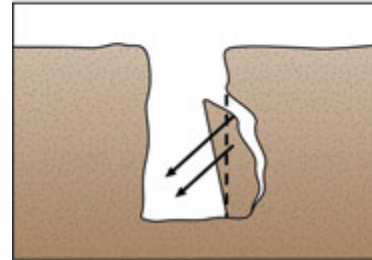
- Only a qualified person can determine if the soil in and around a trench is safe and stable.
- Soils contain a mix of organic material, mineral grains, sand, **silt**, and clay.
- Each of the various soil types, depending on the condition of the soil at the time of the excavation, behaves differently.



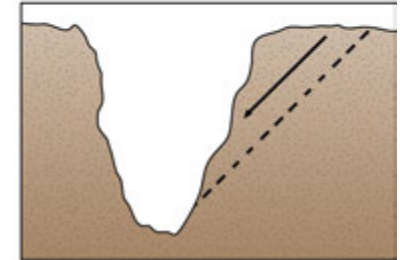
SANDY SOIL COLLAPSES STRAIGHT DOWN



FIRM DRY CLAY AND LOAMS CRACK



WET CLAY AND LOAMS SLAB OFF



WET SANDS AND GRAVELS SLIDE

4.1.0 Soil Composition and Trench Failures (2 of 2)

- When working in or around a trench, stay alert for changes in trench walls that may be an early sign of an impending cave-in.
- OSHA requires a minimum of two feet between the trench wall and the spoils pile in order to prevent soil sliding back into the trench.

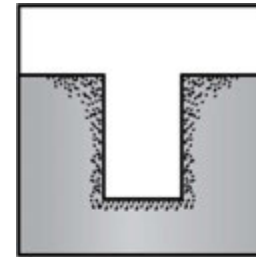


4.1.1 Type A Soil

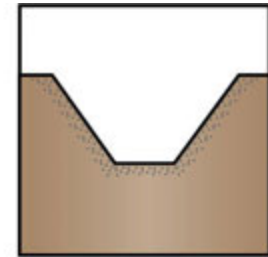
Type A soil has good strength and cohesion, such as clay, silty clay, sandy clay, and clay **loam**.

Soil is not Type A if:

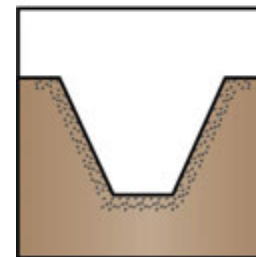
- The soil is fissured.
- The soil can be affected by vibration from heavy traffic, pile driving, or other similar effects.
- The soil has been previously disturbed.



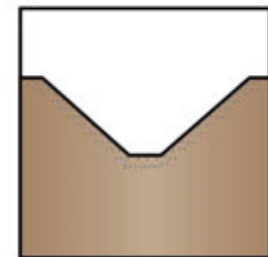
STABLE ROCK
90 DEGREES



TYPE B SOIL
45 DEGREES



TYPE A SOIL
53 DEGREES



TYPE C SOIL
34 DEGREES

4.1.2 Type B Soil

Type B soil refers to cohesive soils with compression strength of greater than 0.5 ton per square foot but less than 1.5 tons per square foot.

Type B soils include:

- Granular soil, including angular gravel (similar to crushed rock)
- Silt
- Sandy loam
- Unstable rock
- Any unstable or fissured Type A soils or previously disturbed soils, except those that would fall into the Type C classification.



4.1.3 Type C Soil

Type C soil is the most unstable soil type. Type C soil refers to cohesive soil with compression strength of 0.5 ton per square foot or less.

Type C soil includes:

- Gravel
- Loamy soil
- Sand
- Any submerged soil or soil from which water is freely seeping
- Unstable submerged rock



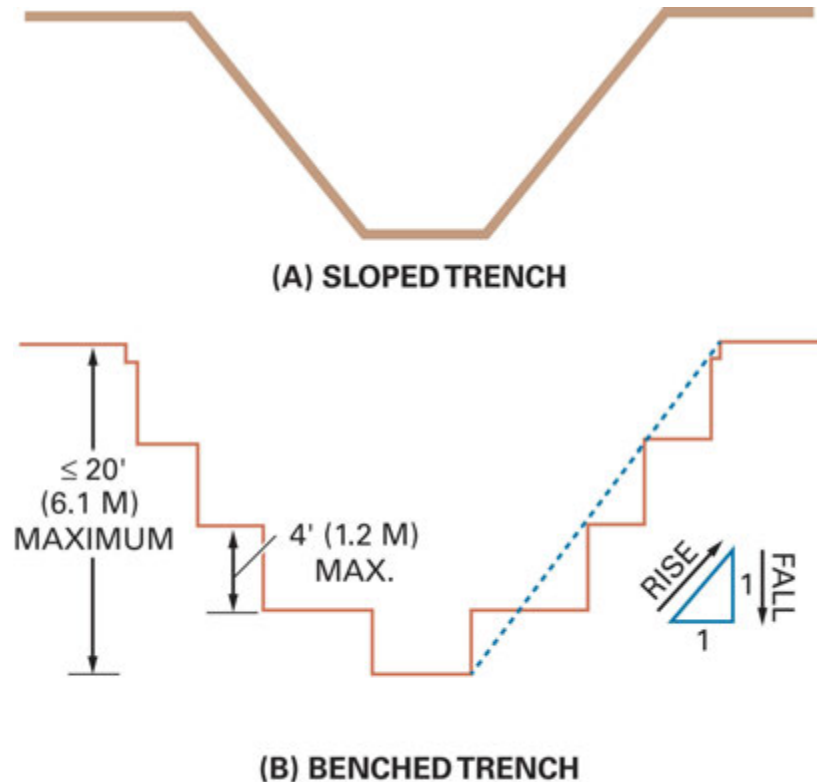
4.2.0 Protecting Workers in Trenches and Excavations

- When working with heavy equipment around any excavation or trench, the operator is responsible for their personal safety, as well as that of others in the work zone.
- A competent person must inspect the site conditions daily and give approval for moving heavy equipment into the excavation area.
- Take care around barricades, open trenches, under loads, and around moving equipment.
- Wear appropriate PPE.
- Make sure sloping, shoring, trench boxes, or benching, are used for excavations and trenches over 5 feet deep.
- Stop work and immediately exit the trench if there is any potential for a cave-in.



4.2.1 Making the Trench Safe (1 of 3)

- Trenches are often reinforced with **shoring** or **shielding** systems.
- Trench walls may also be **sloped**, or cut at an angle away from the trench floor to relieve pressure and avoid cave-ins.



4.2.1 Making the Trench Safe (2 of 3)

Shoring in a trench is placed against the trench walls to support them and prevent their movement and collapse.



4.2.1 Making the Trench Safe (3 of 3)

- Trench shields, also called trench boxes are placed in un-shored trenches to protect personnel from trench wall collapse.
- They are not designed to prevent cave-ins.



4.2.2 Ladders

- Ladders must be placed within 25 feet of any worker.
- Ladder side rails must extend a minimum of 3 feet above the landing.
- Ladders must have nonconductive side rails if used near electricity.
- Two or more ladders must be used where there are 25 or more workers.
- Inspect ladders before each use. Do not use damaged ladders.
- Use ladders only on stable or level surfaces.
- Secure ladders when they are used in any location where they can be displaced by excavation activities or traffic.
- While on a ladder, do not carry any object or load that could cause you to lose your balance.



4.2.3 Sloping Systems

- A sloping system is a method of protecting workers from cave-ins.
- Sloping is done by inclining the sides of an excavation.
- The angle of incline varies according to soil type, environmental conditions of exposure, and nearby loads. These angles are determined by OSHA and MSHA.



4.3.0 Soil-Related Hazards

- Hazards from excavation include soils, dust, microorganisms, and sewage.



4.3.1 Soil and Demolition Dust

- Dust may contain animal droppings, fungi, or other organisms.
- Asbestos dust exists in most demolition work.
- Concrete dust can cause a fatal lung disease called silicosis.

To limit your exposure to soil contaminants and concrete dust:

- Wear a well-fitted respirator or proper PPE.
- Shower and change into clean clothes at your job site when possible.
- Avoid eating or drinking in the dusty area.



4.3.2 Sewage

- Excavation may uncover live sewers or septic tanks.
- Contact with raw sewage can place workers at risk for serious diseases such as E. coli, hepatitis, or pneumonia.
- Wear latex or similar gloves when working around sewage.
- Make sure to tell your employer and see a doctor if you become ill after working on a job site with sewage.



4.3.3 Soil Contamination

- Contaminants can be in the soil, buried underground, or stored in some container such as a barrel.
- Heavy equipment operators are required to report any potential contaminations to a supervisor.
- Notify your supervisor if you find any unusual fluid, foul odors, or unmarked barrels or tanks.



Wrap Up - Trade Terms

Loams: Soils containing clay, sand, and organic matter.

Shielding: The use of structures that are inserted into a trench to protect workers.

Silt: Soil deposited by running water.

Shoring: A support system designed to prevent the trench walls from collapsing.

Sloping: The angling of trench walls in order to reduce the risk of trench collapse.



4.0.0 Section Review Question 1

1. The type of soil that is most likely to slab off is _____.
- a. sand
 - b. dry clay
 - c. wet clay
 - d. gravel



4.0.0 Section Review Question 1 Answer

1. The type of soil that is most likely to slab off is _____.
- a. sand
 - b. dry clay
 - c. wet clay**
 - d. gravel



4.0.0 Section Review Question 2

2. Trenches must be inspected daily by _____.
a. every worker
b. the project manager
c. a supervisor
d. a competent person



4.0.0 Section Review Question 2 Answer

2. Trenches must be inspected daily by _____.
a. every worker
b. the project manager
c. a supervisor
d. a competent person



4.0.0 Section Review Question 3

3. The condition known as silicosis is caused by _____.
- a. coming into contact with sewage
 - b. breathing concrete or cement dust
 - c. handling bird droppings
 - d. coming into contact with rotting vegetation



4.0.0 Section Review Question 3 Answer

3. The condition known as silicosis is caused by _____.
- a. coming into contact with sewage
 - b. breathing concrete or cement dust**
 - c. handling bird droppings
 - d. coming into contact with rotting vegetation



Next...



Module Review and Examination

Complete the Review Questions at the end of the module and prepare for the Module Exam.